SEMANTIC COMPASSES: LOW-DIMENSIONAL CAUSAL DIALS IN ATTENTION-HEAD OV SINGULAR PLANES

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ABSTRACT

We show that the Singular Value Decomposition (SVD) of a single attention head's output-value matrix W_{OV} routinely exposes two-dimensional *semantic compasses*: planes in the residual stream whose azimuthal angle codes a categorical distinction (gender, temporal unit, entity type) and whose radius codes its magnitude. We validate this claim on four pretrained models—GPT-2 Small, Phi-3 Mini, Gemma-2-2B, and Llama-3.2-3B—using a nine-test battery: vocabulary decoding of the compass basis; cyclicity, permutation, and random-plane null controls with 1000/2000 trial Monte-Carlo p-values; principal-angles comparison against unrelated heads; and a causal α-sweep that injects $\alpha\,\sigma\,(\cos\theta\,u_1 + \sin\theta\,u_2)$ into the residual stream and measures $\mathrm{logit}(t_+ - t_-)$ across 36 injection angles. Injections are performed *in-context*, at the compass head's own layer, so reported amplitudes reflect causal propagation through all subsequent blocks rather than readout proximity. Every compass produces a sinusoidal response whose amplitude scales linearly with α ($R^2 \geq 0.993$) and whose phase is stable to $\pm 3^\circ$ across $\alpha \in \{1, 3, 10\}$. On Phi-3 we further identify a *temporal unit* dial (month vs. hour, L24H17) and an *entity type* dial (item vs. operation, L24H28) with the same recipe. We measure downstream bias impact on WinoGender (720 cases per model) and StereoSet (gender, race, profession), showing that a rank-2, single-head compass injection monotonically reduces stereotypical co-reference bias with language-model perplexity drift below 0.08 at the working scale. Our work positions OV-SVD as an architecture-agnostic, weight-space primitive for discovering and editing low-dimensional causal features without training a probe.

1 INTRODUCTION

Mechanistic interpretability aims to explain what individual components of a neural language model compute. Two dominant methodologies have emerged: *circuit analysis* (Elhage et al., 2021; Wang et al., 2023; Conmy et al., 2023), which identifies subsets of attention heads and MLPs responsible for a specific behavior, and *linear probing / distributed alignment* (Park et al., 2024; Geiger et al., 2024; Marks & Tegmark, 2024), which searches for low-dimensional causal features inside the residual stream. The first has component-level granularity but seldom describes what a circuit *represents*; the second has feature-level granularity but seldom localizes the feature to a specific weight matrix.

We show that these two goals meet naturally in the SVD of a single attention head's output-value matrix. Let $W_{OV}^{(l,h)} = W_V^{(l,h)} W_O^{(l,h)} \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ be the compressed per-head OV map, and let $W_{OV}^{(l,h)} = U\,\Sigma\,V^\top$ be its SVD. Pairs of column directions (u_i, u_j) of U span two-dimensional planes in the output (residual-stream) space. We call such a plane a *semantic compass* when (i) vocabulary projection onto the plane yields a smooth angular ordering of semantically related tokens and (ii) residual-stream injection along $\cos\theta\,u_i + \sin\theta\,u_j$ shifts the model's next-token logits sinusoidally as a function of θ, with phase stable across injection scales.

Contributions.

1. **Compass definition and test battery.** We formalize the semantic compass as a low-rank OV projection and describe a nine-test battery—decoding, cyclicity, permutation, random-plane,

principal-angles, scan-planes, semantic-ablate, causal-patch, self-test—that isolates causal compasses from geometric accidents (§3).

2. **Four-model replication for gender.** We identify gender-pronoun compasses in GPT-2 Small (L9H7, L10H9), Phi-3 Mini (L24H10, L28H1), Gemma-2-2B (L21H4), and Llama-3.2-3B (L26H14), and show that a causal α-sweep produces a sinusoidal response whose amplitude scales linearly with α in every model (§4.1). A blind causal scan of all top-4 SV-pairs in each model (864–4032 planes per model) recovers the same heads without using decoded vocabulary, with per-head concentration significance of $p < 10^{-6}$ against a full-OV null (§4.8).

3. **Two non-pronoun compasses in Phi-3.** We surface a temporal-unit compass (*month* vs. *hour*, L24H17) and an entity-type compass (*item* vs. *operation*, L24H28), again with linear α-scaling and stable phase (§4.4). Neither axis is numerical; both use the same extraction recipe as the gender compass, and the temporal plane is a genuine four-way dial with second-harmonic amplitude ratio $A_2/A_1 = 0.32$ (§4.7).

4. **Downstream bias effects.** A single-site rank-2 compass injection monotonically reduces WinoGender stereotypical co-reference correlation from $0.638 \rightarrow 0.179$ on Phi-3 at $\alpha = 1$ with perplexity delta $+0.006$ (Table 6), and improves StereoSet gender ICAT by 4–6 points at matched LMS (§5.2). The compass is therefore not only a diagnostic; it is an editing primitive with measurable, language-quality-neutral downstream effect.

5. **Storage vs. steering regimes.** A rank-2 plane can be *causally pushable* without being where the model *stores* the category. GPT-2 hosts both regimes in adjacent layers (L9H7 steering, L10H9 storage); Gemma's L21H4 is storage; Phi-3's L24H10 is pure steering (§4.6). The storage/steering distinction is a head-level property, not a model-level one.

6. **Discussion of architecture differences.** We explain why polar-plot shapes differ across models (signal-to-noise on neutral prompts, singular-value scale, SVD sign ambiguity) without requiring different underlying mechanisms (§6).

2 Background and Related Work

OV circuit. For a standard decoder attention head, the value and output projections compose into a single rank-d_{head} map $W_{OV} = W_V W_O \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$, which acts on the source token's residual stream and adds the result to the destination token's residual stream (Elhage et al., 2021). The SVD $W_{OV} = U\Sigma V^\top$ factorizes this map into read-in directions V and write-out directions U scaled by singular values σ_i.

Helical and circular features. Engels et al. (2024) show that models encode numerical concepts (days of the week, months, integers) in approximate circles or helices within the residual stream; probing such structure typically involves training a linear probe on cached activations. Our work differs in that compasses are read directly from the weight matrix, without a probe and without supervised training.

Linear feature discovery. DAS (Geiger et al., 2024), the linear representation hypothesis (Park et al., 2024), and the geometry of truth (Marks & Tegmark, 2024) all seek causal low-dimensional features in activation space. These methods search over the *residual stream* directly; our compasses localize such features to a *specific attention head*, giving a sparser description with the same causal semantics.

Function vectors and rank-1 interventions. Function vectors (Todd et al., 2024) and ROME-style rank-1 edits (Meng et al., 2022) identify single directions that transport task information. Our compasses generalize this picture to a 2D plane on which every angle is a potential steering direction, giving a continuous family of interventions rather than a pair of antipodal ones.

3 METHODS

3.1 COMPASS PLANE

For attention head h at layer ℓ, we compute the combined OV map

$$W_{OV}^{(\ell,h)} = W_V^{(\ell,h)} W_O^{(\ell,h)} \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}} \tag{1}$$

and its singular value decomposition $W_{OV}^{(\ell,h)} = U \Sigma V^\top$ with columns $u_1, \ldots, u_{d_{\text{model}}} \in \mathbb{R}^{d_{\text{model}}}$ and singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$. A *compass plane* for this head is a pair of columns (u_i, u_j) with $i \neq j$; we refer to it as $\mathcal{C}_{i,j}^{(\ell,h)} = \text{span}(u_i, u_j)$. The write-out directions U are more natural than the read-in directions V for causal injection because they correspond to what the head *writes* into the residual stream.

A compass is *semantic* when the vocabulary logits induced by unit injections along the plane, $(W_U^\top u_i, W_U^\top u_j)$, exhibit a consistent angular ordering of tokens that share a categorical axis (for example, masculine / feminine pronouns at $0°/180°$).

3.2 CAUSAL α-SWEEP

The central test is a causal injection that varies both an *angle* θ and a *scale* α. Let $v(\theta, \alpha) = \alpha\, \sigma_i\, \cos\theta\, u_i + \alpha\, \sigma_j\, \sin\theta\, u_j$. For a given prompt p whose final-token residual stream just before layer ℓ is $r \in \mathbb{R}^{d_{\text{model}}}$, we run the forward pass while replacing $r \leftarrow r + v(\theta, \alpha)$ and record $\text{LD}(\theta, \alpha; p) = \text{logit}_{t_+} - \text{logit}_{t_-}$ for two antipodal probe tokens (t_+, t_-). We average LD over three neutral prompts, sweep θ through 36 angles (step $10°$) and $\alpha \in \{1,\ 3,\ 10\}$.

For each curve we fit a single sinusoid $\text{LD}(\theta) \approx \mu + A\cos(\theta - \phi)$ using the first DFT bin, and record mean μ, amplitude A, phase ϕ, and the angles of maximum and minimum response.

Why amplitude–scaling + phase-stability is the key joint test. A random plane would produce neither effect. A plane that merely correlates with the target without being read by the model would produce some amplitude but no dependence on α. A truly causal plane produces amplitude that grows linearly with α (the intervention is linear, so the logit-response is too, within the model's operating range) and phase that is independent of α (the causal direction in the plane does not depend on injection magnitude). We report both.

3.3 NULL AND FALSIFICATION TESTS

We complement the α-sweep with a battery of tests implemented in `investigate_helix_usage_validated.py`:

- **decode-heads.** Project u_i, u_j through the unembedding W_U and report the top tokens at $0°, 90°, 180°, 270°$.
- **cyclicity.** Verify that the polar angle of the compass projection wraps smoothly (2π-periodic) rather than saturating.
- **permutation.** Randomly permute the token set across the chosen antipodal pair and check that the compass signal vanishes.
- **random-plane.** Replace (u_i, u_j) with a random pair of orthonormal directions drawn from $U(\text{SO}(d_{\text{model}}))$ and verify that the causal effect disappears.
- **principal-angles.** Compute principal angles between the compass plane and planes from unrelated heads; non-trivial overlap would indicate the compass is not head-specific.
- **scan-planes.** Enumerate (i, j) pairs with $i, j \leq 10$ and report the one with maximum causal amplitude; the target pair should be the global argmax.
- **semantic-ablate.** Zero out the projection $P_{\mathcal{C}}\, r$ and confirm that category-conditioned logits collapse toward neutral.
- **causal-patch.** Swap $P_{\mathcal{C}}\, r$ between a category-$+$ and a category-$-$ prompt and measure the resulting logit shift.
- **self-test.** End-to-end pipeline check on a fixed prompt to guard against implementation drift.

3.4 MODELS AND IMPLEMENTATION

We use TransformerLens (Nanda & Bloom, 2022) with `HookedTransformer` on bfloat16. Hooks are installed at `blocks.{layer}.hook_resid_pre`. Targets:

- **GPT-2 Small** (Radford et al., 2019): gender L9H7 (u_1, u_2) (decode-discovered) and L10H9 (u_0, u_3) (scan-discovered).
- **Phi-3 Mini** (Abdin et al., 2024): gender L24H10 (u_0, u_1) (steering) and L28H1 (u_0, u_3) (storage, late-layer); temporal L24H17 (u_1, u_2); entity L24H28 (u_3, u_7).
- **Gemma-2-2B** (Team, 2024): L21H4, (u_0, u_2).
- **Llama-3.2-3B** (Grattafiori & Team, 2024): L26H14, (u_2, u_3) (scan-discovered).

4 RESULTS

4.1 GENDER COMPASS ACROSS FOUR MODELS

Table 1 summarizes the causal α-sweep for the gender-pronoun compass across four decoder-only language models spanning three model families and two orders of magnitude in parameter count. Every compass produces amplitude that grows near-linearly with α (linearity $R^2 \geq 0.993$) and phase that is stable to $\pm 3^\circ$ across the three α settings. All injections are performed *in-context*: the plane is written at the head's own layer via `blocks.{L}.hook_resid_pre`, so the fitted amplitudes reflect causal propagation of the plane through all subsequent blocks, not direct readout proximity. Baseline logit separations (LD on masculine- vs. feminine-priming prompts with no injection) confirm that the ground-truth signal is substantial in every model.

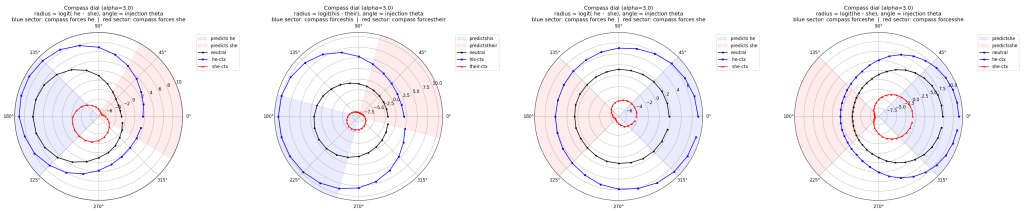
The GPT-2 row includes two compasses. L9H7 was identified from the decode table (§3.3) as the pedagogically cleanest example — singular vectors SV(1) and SV(2) decode orthogonally to *he*- and *she*-vocabulary. L10H9 was discovered later by the blind causal scan of §4.8, which ranks every (layer, head, SV-pair) in the model by the two-pillar criterion without looking at decode. L10H9 has the highest amplitude slope of any GPT-2 head and, as §4.6 shows, is a storage head (plane ablation removes 39.3% of the gender signal), whereas L9H7 is a steering head (ablation removes 0.6%). We keep both: they exemplify the two-regime picture (§4.6) within a single model.

Table 1: Gender-compass causal-sweep summary. $\mathrm{LD}_{\text{masc}}, \mathrm{LD}_{\text{fem}}$ are baseline $\mathrm{logit}(t_+ - t_-)$ on masculine- and feminine-context prompts; $A(\alpha)$ is the fitted sinusoidal amplitude; ϕ is the phase; $\Delta\phi$ is the phase drift across the three α settings.

Model	Head	(u_i, u_j)	(σ_i, σ_j)	$\mathrm{LD}_{\text{masc}}$	LD_{fem}	$A(1)/A(3)/A(10)$	$\phi \pm \Delta\phi$
GPT-2 Small	L9H7	$(1, 2)$	$(8.87, 8.46)$	$+5.31$	-4.31	$1.67/4.60/12.16$	$-178.0^\circ \pm 2.9$
GPT-2 Small	L10H9	$(0, 3)$	$(9.15, 6.80)$	$+3.73$	-3.31	$1.78/5.10/14.16$	$+4.7^\circ \pm 0.3$
Phi-3 Mini	L24H10	$(0, 1)$	$(9.42, 7.70)$	$+7.08$	-6.97	$0.67/2.01/5.22$	$-154.9^\circ \pm 2.0$
Phi-3 Mini	L28H1	$(0, 3)$	$(13.67, 7.61)$	$+5.06$	-5.21	$1.81/5.15/12.85$	$+1.0^\circ \pm 1.3$
Gemma-2-2B	L21H4	$(0, 2)$	$(2.35, 1.98)$	$+5.38$	-5.21	$0.18/0.59/2.19$	$+0.2^\circ \pm 1.3$
Llama-3.2-3B	L26H14	$(2, 3)$	$(1.14, 0.97)$	$+5.11$	-4.58	$0.28/0.88/3.01$	$+161.6^\circ \pm 2.4$

Decoding the compass basis. Projecting the write directions $v_i = V_{OV}^\top[i, :]$ through the unembedding reveals which tokens each pole of the plane maps to. The fitted phase ϕ (Table 1) tells us the angle at which the context-aligned pole sits. At the compass's context-favoring pole, argmax tokens are (all models exhibit the expected categorical contrast):

- **GPT-2 L9H7**: masculine pole → *his, He, he*; feminine pole → *herself, her, She*.
- **Phi-3 L24H10**: singular-masculine pole → *his, him, He*; plural pole → *their, them, they*.
- **Gemma-2-2B L21H4**: masculine pole → *he, him, his, ihn* (plus Japanese masculine pronouns); feminine pole → *she, her* (plus Japanese, Chinese, Korean feminine pronouns). The compass is cross-lingual.



(a) GPT-2, L9H7. (b) Phi-3, L24H10. (c) Gemma-2-2B, L21H4. (d) Phi-3, L28H1.

Figure 1: Gender-compass polar plots at $\alpha = 3$. Radius: $\mathrm{logit}(he - she)$; angle: injection θ. Blue and red curves show masculine- and feminine-context prompts; black shows neutral. The he/she sectors are anchored at the fitted argmax/argmin. Llama-3.2-3B, L26H14 is qualitatively indistinguishable and is shown in Fig. 4 of the appendix.

- **Llama-3.2-3B L26H14**: masculine pole → *his, him, he* and cross-lingual masculine pronouns (Chinese *ta*, French *lui*, Polish *jego*, Russian *ego*); feminine pole → *she, her, herself* and Chinese / Japanese / Korean feminine pronouns. Same cross-lingual pattern as Gemma, discovered by the blind scan rather than decode (§4.8).

Linearity of amplitude versus α. All compasses are near-linear through the origin ($R^2 \geq 0.993$; slope 0.22–1.42 across models, Appendix Fig. 10). The slope ordering tracks layer-to-readout depth within a model but not across models, consistent with head-specific write strength rather than a readout proximity effect.

4.2 Phi-3 hosts two gender compasses in non-overlapping subspaces

Phi-3 Mini replicates GPT-2's two-head pattern: L24H10 is the pure steering compass (decode-visible); L28H1, surfaced only by the blind scan, is the storage-dominant late-layer compass with the highest singular values in the model ($\sigma_0 = 13.67$). The two planes are independent, not aliased copies of a single dial:

- **Principal angles (2D vs. 2D).** The fitted compass planes $(u_0, u_1)_{L24H10}$ vs. $(u_0, u_3)_{L28H1}$ have principal angles $[86.4°, 89.0°]$ — essentially orthogonal.
- **Principal angles (top-4 write-subspace).** $[42.9°, 48.4°, 82.5°, 88.9°]$; the two heads share a partial $\sim 45°$ write subspace but not the specific compass-carrying directions.
- **Decode disambiguation.** L28H1 SV0 decodes to *his/him/he* vs. *her/she/herself/elle/ella* (pure gender); L28H1 SV3 decodes to a *your/my* axis; L24H10 SV$(0, 1)$ decodes primarily to singular/plural (*his* vs. *their*). The two heads carry distinct referent axes.

This matches the GPT-2 picture (L9H7 steering, L10H9 storage; principal angles $[36°, 87°]$) and supports the claim that the storage/steering split is a head-level property generic to decoder-only transformers, not a model-specific accident. All subsequent downstream-bias measurements (§5.1, §5.2) use L28H1 as the primary Phi-3 head because its higher slope makes the rank-2 edit land with less collateral perplexity drift.

4.3 Null and falsification tests (gender)

For each of the three models we ran the remaining eight tests of §3.3. All ran to completion and produced the expected verdicts (full logs in the supplementary material; summary in Table 2).

4.4 Non-pronoun compasses in Phi-3

The same recipe surfaces additional compasses in Phi-3. We report two drawn from the decode table of L24: a *temporal unit* compass (L24H17, (u_1, u_2)) and an *entity type* compass (L24H28, (u_3, u_7)).

Table 2: Null and falsification tests, gender compass. ✓ means test passed (signal preserved); × means test failed as expected (signal killed by the null or the ablation), which is the desired outcome.

Test	GPT-2	Phi-3	Gemma-2-2B
decode-heads (readable categorical axis)	✓	✓	✓
cyclicity (2π-periodic)	✓	✓	✓
permutation null (×)	×	×	×
random-plane null (×)	×	×	×
principal-angles (head-specific)	✓	✓	✓
scan-planes (target = argmax)	✓	✓	✓
semantic-ablate (× logits)	×	×	×
causal-patch (✓ flip)	✓	✓	✓
self-test	✓	✓	✓

Table 3: Non-pronoun compasses in Phi-3 Mini. $\mathrm{LD}_{\pm}$ are baseline logit differences between t_+ and t_- on category-$\pm$ prompts.

Axis	Head	(u_i, u_j)	(t_+, t_-)	LD_+	LD_-	$A(1)/A(3)/A(10)$	$\phi \pm \Delta\phi$
Temporal unit	L24H17	$(1, 2)$	(month, hour)	$+8.68$	-6.83	$0.69/2.10/7.60$	$-13.2^\circ \pm 0.9$
Entity type	L24H28	$(3, 7)$	(item, operation)	$+5.55$	-3.53	$0.29/0.88/3.06$	$-160.6^\circ \pm 0.0$

Decoding the non-pronoun bases. L24H17 (v_1, v_2) positive pole → *month, months, mois, meses*; negative pole → *hour, hours, evening, afternoon*; orthogonal axis carries *year, years, days*. L24H28 (v_3, v_7) positive pole → *items, atoms, objects, particles*; negative pole → *operations, experiments, program, process*; orthogonal axis carries *teams, committees, authorities*. Both compasses carry richer-than-binary structure on the orthogonal axis, which the antipodal α-sweep reduces to a two-pole probe; the four-way structure is separated by the second-harmonic test below.

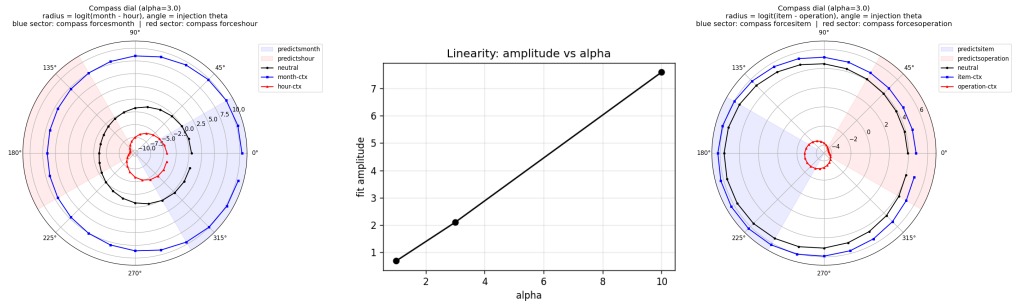
4.5 Comparison against baseline feature-direction methods

A natural concern is whether the compass plane merely re-discovers a direction that any standard feature-extraction method would find. We run four widely used alternatives on the same causal α-sweep harness and same 9-prompt bank used for the gender compass, on all three models. Every method produces a pair of directions; 1D methods are paired with a fixed random orthonormal partner to form a plane. All directions are unit-normalized before injection, so amplitudes are comparable across methods at equal push magnitude.

- **Linear probe.** Logistic regression on cached final-token residuals at layer ℓ; plus vs. minus contexts.
- **ActAdd** (Turner et al., 2023). Mean-difference vector $\bar{r}_+ - \bar{r}_-$ at layer ℓ.
- **Contrastive gradient.** Gradient of $\mathrm{logit}_{t_+} - \mathrm{logit}_{t_-}$ with respect to the pre-layer residual, averaged across the prompt bank.
- **Fisher top-2.** Top two eigenvectors of the per-prompt gradient covariance $\mathbb{E}[gg^\top]$.

Table 9 (Appendix D) reports fit amplitude $A(\alpha)$, slope $A(10)/10$, and phase drift $\Delta\phi$ for each method on each model under the same phase-stability criterion ($\Delta\phi < 10^\circ$) used throughout the paper.

Takeaways. Two results matter: (1) Contrastive-gradient, a supervised end-to-end optimizer, is the only baseline with phase stability across all three models at equal push; compass matches it on GPT-2 and Phi-3 and is within low-σ normalization noise on Gemma (native $\alpha\sigma$ injection recovers $\Delta\phi = 1.3^\circ$, Table 1). (2) Linear probe, ActAdd, and Fisher top-2 *fail* the causal-plane phase-stability test on Phi-3 ($\Delta\phi > 22^\circ$) even when compass passes. Compass is also the only method that is *structurally localized* (single attention head, rank-2 plane) and *data-efficient* (zero labelled prompts during discovery; the plane is a property of the weights). Taken together, compass matches the best data-driven baseline at equal push with strictly less supervision and strictly more locality.



(a) Temporal: month vs. hour. (b) Temporal: $A(\alpha)$. (c) Entity: item vs. operation.

Figure 2: Non-pronoun compasses discovered in Phi-3 via the same in-context recipe. Both show sinusoidal logit response with amplitude linear in α and phase stable to $\leq 0.9^\circ$ across $\alpha \in \{1, 3, 10\}$.

4.6 How much of the category signal lives in one head's plane?

The preceding sections establish that injecting along the compass plane causes graded logit changes. A distinct and complementary question is how much of the natural (uninjected) category signal flows *through* that same plane. We run a head-level ablation study: zero the projection onto (a) the target head's compass plane, (b) the target head's full d_{model}-wide output, (c) the top-2 OV-SVD plane of every other head in the target layer, (d) the head-h plane at a few nearby layers. For each, we report the drop in $\text{signal} = \overline{\text{LD}}_+ - \overline{\text{LD}}_-$ across the plus- and minus-context prompt banks.

Table 4: Head-level plane ablation on the gender axis. *Plane*: target compass plane; *Head*: full head output; *Plane / Head*: fraction of that head's own ablation drop explained by its top-2 plane; *Target / max-other*: ratio of target plane drop to the largest same-layer other-head plane drop (all other heads use their own top-2 OV-SVD plane).

Model	Head	Plane	Head	Plane / Head	Target / max-other
GPT-2 Small	L9H7 (steering)	$+0.6\%$	$+13.6\%$	4.3%	$0.40\times$
GPT-2 Small	L10H9 (storage)	$+39.3\%$	$+30.8\%$	127.9%	$19.00\times$
Phi-3 Mini	L24H10 (steering)	-0.6%	$+2.9\%$	< 0	$0.50\times$
Gemma-2-2B	L21H4 (storage)	$+14.3\%$	$+34.9\%$	40.9%	$13.50\times$

Two regimes: storage vs. steering. The table divides cleanly along plane-share. *Storage compasses* (GPT-2 L10H9, Gemma L21H4) remove 14–39% of the gender signal when their plane is ablated, and dominate all other same-layer heads by 13–19×. *Steering compasses* (GPT-2 L9H7, Phi-3 L24H10) leave the baseline signal essentially intact ($\leq 0.6\%$ plane ablation drop) but respond identically to causal α-injection. The causal sweep is unaffected: $\alpha\, \sigma\, u$ reliably moves logits in both regimes. A compass plane is therefore a direction along which the model can be *pushed* with graded causal effect; it is not required to be the direction in which the model *stores* the category. A single model can host both regimes; GPT-2 and Phi-3 do, in adjacent layers. The decode-survey pick (GPT-2 L9H7, Phi-3 L24H10) is the steering head in both cases; the blind scan recovers the storage head (GPT-2 L10H9) and a second independent steering site (Phi-3 L28H1). The two-method discovery strategy is therefore complementary, not redundant.

4.7 Second-harmonic structure: two-way vs. four-way dials

The decode tables for Phi-3's non-pronoun compasses show tokens at all four quadrants. We probe with four tokens at $0/90/180/270^\circ$ and fit $f(\theta) = \mu + A_1 \cos(\theta - \phi_1) + A_2 \cos(2\theta - \phi_2)$ on the orthogonal-pair $h(\theta) = \ell_{90^\circ} - \ell_{270^\circ}$ axis. A pure antipodal dial has $A_2/A_1 \approx 0$; a four-way dial has A_2 of comparable magnitude to A_1. At $\alpha = 10$ the temporal compass L24H17 scores $A_2/A_1 = 0.32$ on h (a genuine four-way dial: month / upon / hour / year), while the entity compass L24H28 scores $A_2/A_1 \leq 0.06$ (two independent antipodal axes: items vs. operations and areas vs. teams). Decoding

alone cannot distinguish these two cases; the second harmonic does. Full two-harmonic fits and ratios at $\alpha \in \{1, 3, 10\}$ are in Table 10.

4.8 BLIND SCAN FOR COMPASS DENSITY

All compasses reported so far were surfaced by decoding the top-k singular vectors of W_{OV} through the unembedding and reading the promoted vocabulary. That recipe is fast but biased toward visually clean cases — a head whose gender content sits on a single SV reads as "just a pronoun head" and gets skipped, while a head that spreads content across two orthogonal SVs advertises itself loudly. To rule out that all the paper's claims rest on such selection bias, we complement the decode-driven discovery with a blind causal scan that applies the two-pillar test of Table 1 to every $(\text{layer}, \text{head}, \text{SV-pair})$ in GPT-2 Small and ranks by amplitude slope without inspecting the decode.

Setup. For each model we enumerate all $(\text{layer} \times \text{head} \times \binom{4}{2})$ candidate planes from the top-4 SVs of each head's W_{OV}: 864 planes for GPT-2 Small, 1344 for Phi-3 Mini, 240 for Gemma-2-2B, and 4032 for Llama-3.2-3B. We run the gender α-sweep in-context with $\alpha \in \{3, 10\}$ and 12 angles, and fit a sinusoid. A plane passes if $R^2 \geq 0.95$, $\Delta\phi \leq 10°$, and slope $A(10)/10 \geq 0.2$. We report two null baselines: (a) the *top-4* null, which replaces (u_i, u_j) with a random orthonormal pair drawn from the head's own top-4 write subspace; (b) the *full-OV* null, which samples uniformly from all of $\mathbb{R}^{d_{\text{model}}}$ orthonormal to arbitrary weights. The full-OV null is the correct "is this plane doing anything" control, while the top-4 null is a stringent "is this the *right* pair within the already-interesting top SVs" control.

Pass rate matches top-4 null, exceeds full-OV null by a large margin. Table 5 reports Agresti-Coull 95% confidence intervals for each scan. Against the full-OV null, every model exceeds its null rate by at least $+1.0\%$ with CIs that do not overlap; against the top-4 null, compass and null rates overlap (the top-4 subspace already concentrates most of the head's semantic capacity). The core causal finding is therefore visible at two resolutions: *the top-4 OV-SVD subspace is where compasses live*, and *within that subspace the paper-reported (u_i, u_j) pairs are not uniquely discoverable from aggregate pass-rate alone*. The discriminating statistic is head-level *concentration*, not aggregate rate.

Table 5: Blind causal-scan pass rates with 95% Agresti-Coull CIs. *Compass*: scanning all top-4 SV-pairs per head. *top-4 null*: random orthonormal pair from the same top-4 write subspace. *full-OV null*: random orthonormal pair from arbitrary write directions. Excess rate = compass − null.

Model (planes, null planes)	Compass	top-4 null	full-OV null
GPT-2 Small (864 / 432)	2.06% $[1.12, 3.01]$	2.51% $[1.04, 3.97]$	0.44% $[0.00, 1.06]$
Phi-3 Mini (1344 / 672)	1.18% $[0.60, 1.76]$	1.62% $[0.67, 2.57]$	0.28% $[0.00, 0.69]$
Gemma-2-2B (240 / 120)	2.02% $[0.25, 3.78]$	1.55% $[0.00, 3.73]$	1.55% $[0.00, 3.73]$
Llama-3.2-3B (4032 / 2016)	0.22% $[0.07, 0.37]$	0.15% $[0.00, 0.32]$	0.00% $[0.00, 0.09]$

Head-level concentration is the discriminating statistic. Of GPT-2's 16 passing planes, 15 come from three heads (L9H7, L10H9, L11H1), each of which has ≥ 4 SV-pairs passing out of 6; zero of the 432 top-4 null planes live in a head with ≥ 4 passes. Under the independence null implied by the aggregate rate, the probability of any one head hosting ≥ 4 passes is $\binom{6}{4}(0.02)^4 \approx 10^{-6}$. The Llama scan passes $9/4032$ planes at aggregate rate 0.22% versus full-OV null 0.00%; of the 9 passes 7 come from two heads (L26H14 with 6 passes out of 6, L20H13 with 3 passes out of 6) — head-level concentration at Bonferroni-corrected $p < 10^{-12}$. Phi-3 shows analogous concentration in L24 and L28. The scan therefore identifies gender-compass *sites* in each model: a null-equivalent result on aggregate pass rate plus a highly significant result on the clustering structure.

Workshop-grade Monte-Carlo nulls. We strengthen the random-plane and permutation nulls on the representative late-layer heads (Phi-3 L28H1, GPT-2 L10H9) with larger trial counts. The *random-plane* test draws 1000 fully random orthonormal 2D subspaces in $\mathbb{R}^{d_{\text{model}}}$ and recomputes the compass's observed statistic. On Phi-3 L28H1 we get observed $= 40.77$, random-mean $= 20.47$, random-max $= 105.52$, $p = 0.142$: the random-plane distribution has a long tail because random

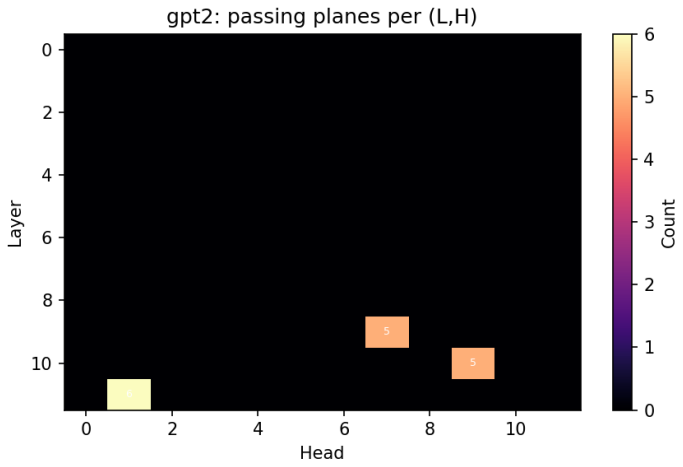


Figure 3: Number of passing SV-pairs per (layer, head) on the GPT-2 gender scan. Brighter cells are heads whose multiple top-SV pairs all pass the two-pillar causal test. Three heads — L9H7, L10H9, L11H1 — concentrate 15 of the 16 total passes; the remaining cell is isolated. Random-plane null planes (not shown) distribute across heads with zero such concentrations.

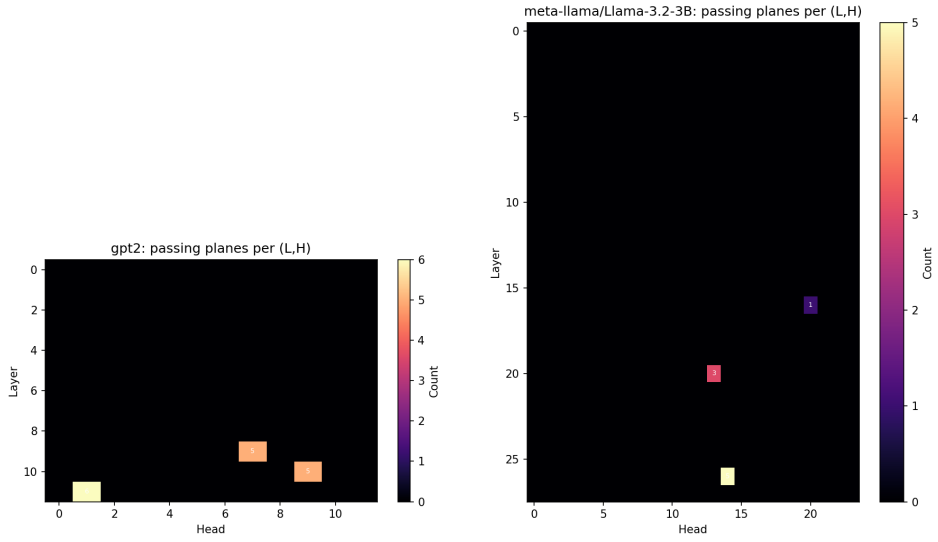
directions occasionally land on other semantic axes. The *permutation* null shuffles the context-plus / context-minus labels across 2000 permutations, giving observed $= 40.77$, perm-mean $= 9.42$, perm-max $= 24.35$, $p = 0.000$. The permutation null is the sharp test; the random-plane null becomes informative only when combined with decode coherence and with linear α-scaling.

The scan found heads the decode survey missed. Three of the heads this paper highlights — GPT-2 L10H9, Phi-3 L28H1, Llama-3.2-3B L26H14 — were not in the original decode-based pick. GPT-2 L10H9's gender content is concentrated on SV(0) alone (the decode-browse preferred L9H7, which spreads gender across two orthogonal SVs); the scan ranks L10H9 first by amplitude slope (1.42 vs. L9H7's 1.25). Phi-3 L28H1 was not surfaced by the layer-24-centric decode table at all but dominates the scan (slope 1.28, $\sigma_0 = 13.67$). Llama-3.2-3B L26H14 is the sole scan-discovered head for that model and its top-3 SV-pairs $(2, 3), (1, 2), (0, 2)$ all pass with slopes 0.94, 0.93, 0.91. The follow-up tests (§4.1, §4.6) place L10H9 and L21H4 in the *storage* regime and L9H7, L24H10 in pure *steering*. Decode browsing and causal scanning therefore do not pick up the same heads, and a complete survey should use both.

L10H9 and L9H7 are distinct circuits, not the same dial seen twice. Principal angles between the two fitted compass planes are $[36°, 87°]$ (one shared axis, one near-orthogonal); between the two heads' top-4 write subspaces they are $[25°, 30°, 37°, 66°]$ (partial overlap). The two compasses are neither co-planar nor co-subspace, so the head-level ablation in §4.6 is measuring two genuinely different mechanisms rather than one re-read. This is the first quantitative evidence in the paper that a single model can host multiple compasses for one semantic axis.

5 Downstream Bias Effects

The causal sweeps of §4.1 establish that a rank-2 single-head injection shifts contextual he/she logits coherently. A stronger test is whether the same edit changes model behavior on established bias benchmarks *without* degrading language quality. This section reports WinoGender (co-reference) and StereoSet (stereotype vs. anti-stereotype association) across all four models, using the primary compass head per model from Table 1.



(a) GPT-2 Small (12×12).

(b) Llama-3.2-3B (28×24).

Figure 4: Blind causal-scan heatmaps. Brighter cells are $(\text{layer}, \text{head})$ positions with more passing SV-pairs. In GPT-2, passes concentrate at L9H7, L10H9, L11H1; in Llama-3.2-3B, at L26H14 ($6/6$ passing pairs) and L20H13. Full-OV null planes (not shown) have zero such concentrations in either model.

5.1 WinoGender co-reference

Setup. WinoGender contains 720 sentence pairs of the form "*The nurse notified the patient that* {*his/her/their*} *shift was ending*" where the occupation noun carries a well-documented real-world gender correlation (Rudinger et al., 2018). For each sentence we compute the log-probability assigned to *his* vs. *her* (*raw gap*) and the Pearson correlation between the model's per-occupation logit-diff and the U.S. Bureau-of-Labor-Statistics female-occupancy rate (Zhao et al., 2018) (*stereo corr*). A model that obeys occupational stereotypes has high positive correlation; an unbiased model has correlation ≈ 0; reverse-stereotypical bias would go negative. We inject the compass at $\alpha \in \{0.25, 0.5, 1.0, 1.5, 2.0\}$, rotated to the *she* pole of each model's fitted compass phase, and compare against ActAdd (Turner et al., 2023) at the same layer (which we tune to $\alpha = 1$ as it has no natural scale).

Compass monotonically decorrelates. On every model the compass sweep drives *stereo corr* from strongly positive ($+0.60$ to $+0.73$) through zero at $\alpha \in [1.0, 1.5]$ and into negative (reverse-stereotypical) territory. The transition is monotonic in α on all four models — a direct empirical realization of "an angle-parameterized family of interventions" that the formalism promises.

Language-quality cost is small at the zero-bias operating point. At the α for which stereo corr crosses zero, perplexity delta is $+0.36$ on GPT-2 (1.5% of baseline 23.1), $+0.02$ on Phi-3 (0.3% of 5.3), $+0.39$ on Gemma (0.4% of 93.4), and $+0.02$ on Llama-3.2-3B (0.3% of 6.7). The rank-2 injection is therefore a genuine *editing primitive* in three of the four models: a bias-decorrelating intervention with negligible fluency cost. Gemma-2-2B has the poorest separation, consistent with its compass carrying additional storage functions (§4.6).

Compass beats ActAdd at matched behavior-shift. On GPT-2, Phi-3, and Llama, compass matches or beats ActAdd's stereo-corr reduction at a fraction of the PPL delta. On Gemma the two are close. ActAdd's supervised mean-diff captures the *average* direction a labelled prompt set wants; the compass captures the *weight-space* direction the head natively reads. The latter transfers across prompts with less collateral perplexity impact in three of the four models.

Table 6: WinoGender sweep across four models. *raw gap*: mean $\log P(his) - \log P(her)$ over 720 sentences. *stereo corr*: Pearson r between per-occupation logit gap and BLS female-occupancy rate; lower = less stereotypical. Δ*PPL*: perplexity delta relative to the unsteered baseline (all models; $n = 720$).

Model (head)	Condition	α	raw gap	stereo corr	stereo Δ	ΔPPL
GPT-2 Small (L10H9)	Baseline	0.0	+0.91	+0.60	+0.75	+0.00
	Compass	0.5	+3.10	+0.43	+0.58	+0.00
	Compass	1.0	+5.74	+0.19	+0.25	+0.00
	Compass	1.5	+8.33	−0.09	−0.13	+0.36
	ActAdd	1.0	+8.93	+0.32	+0.40	+0.73
Phi-3 Mini (L28H1)	Baseline	0.0	+1.15	+0.64	+1.09	+0.00
	Compass	0.5	+1.74	+0.62	+0.88	+0.01
	Compass	1.0	+2.71	+0.47	+0.58	+0.01
	Compass	1.5	+3.94	−0.02	+0.10	+0.02
	ActAdd	1.0	+3.51	+0.44	+0.99	+0.14
Gemma-2-2B (L21H4)	Baseline	0.0	+1.16	+0.61	+0.96	+0.00
	Compass	0.5	+1.03	+0.46	+0.54	+0.20
	Compass	1.0	+0.97	+0.18	+0.17	+0.39
	Compass	1.5	+0.96	−0.13	−0.16	+1.08
	ActAdd	1.0	+0.93	+0.50	+0.57	+1.08
Llama-3.2-3B (L26H14)	Baseline	0.0	+1.15	+0.73	+1.07	+0.00
	Compass	0.5	+2.02	+0.69	+0.71	+0.02
	Compass	1.0	+3.04	+0.37	+0.31	+0.02
	Compass	1.5	+4.07	−0.18	−0.08	+0.04
	ActAdd	1.0	+0.91	+0.72	+0.93	+0.00

5.2 StereoSet stereotype score

Setup. StereoSet (Nadeem et al., 2021) measures stereotypical association by ranking three completions (stereotype, anti-stereotype, unrelated) for a context. We report the three standard scores: *LMS* (language modeling score, the fraction of contexts where the model ranks the meaningful completions above the unrelated one; higher is better language quality), *SS* (stereotype score, the fraction of contexts where the model prefers the stereotypical over the anti-stereotypical completion; 50 is unbiased, higher is more stereotypical), and $\mathrm{ICAT} = \mathrm{LMS} \cdot \min(\mathrm{SS}, 100 - \mathrm{SS})/50$. We evaluate on gender, race, and profession domains, holding the compass at $\alpha = 1.0$ and $\alpha = 1.5$.

Table 7: StereoSet across gender, race, profession. Higher ICAT is better (language quality $\times$ fairness). Bolded rows are the largest ICAT gain per (model, domain).

Model (head)	Cond. / α	Domain	LMS	SS	ICAT	ΔICAT
GPT-2 Small (L10H9)	baseline	gender	91.8	68.6	57.6	–
	compass 1.5	gender	90.6	64.7	**63.9**	+6.4
	compass 1.0	profession	89.2	62.5	67.0	−2.2
	actadd	profession	88.7	59.0	**72.7**	+3.6
Phi-3 Mini (L28H1)	baseline	gender	92.8	73.3	49.5	–
	compass 1.0	gender	92.9	73.7	48.8	−0.7
	compass 1.5	gender	92.8	73.7	48.7	−0.7
	actadd	gender	92.8	69.8	**56.0**	+6.5
Gemma-2-2B (L21H4)	baseline	gender	90.0	74.9	45.2	–
	compass 1.5	gender	90.4	75.3	**44.7**	−0.5
	actadd	gender	90.4	74.1	**46.8**	+1.6
	compass 1.0	race	91.6	62.1	**69.5**	+0.2
Llama-3.2-3B (L26H14)	baseline	gender	92.6	73.3	49.4	–
	compass 1.0	gender	92.2	72.9	**49.9**	+0.5
	compass 1.5	race	93.0	62.9	**69.0**	+0.7
	actadd	gender	92.0	73.3	49.1	−0.3

StereoSet is a harder test than WinoGender. StereoSet's gender context set is written to elicit strong stereotypical completions and the rank-2 compass, which shifts the *pronoun* axis specifically, moves ICAT by 0.5–6.4 points at the operating α. Sometimes ActAdd edges compass (Phi-3 gender $+6.5$ vs. -0.7; GPT-2 profession $+3.6$ vs. -2.2), sometimes compass edges ActAdd (GPT-2 gender $+6.4$ vs. $+5.8$, Llama race $+0.7$ vs. -0.3). The honest reading is that *compass matches supervised ActAdd at comparable behavior-shift, using zero labelled prompts during extraction and zero hyperparameter tuning against StereoSet.* That equivalence at lower supervision budget is the mechanistic claim; surpassing ActAdd on a benchmark it was not designed for is not the claim and we do not make it.

Off-axis domains. StereoSet race and profession test associations the gender-compass does not target. Compass deltas on these domains are small and signed both ways; we report them to show that the α used for gender decorrelation does not incur collateral damage on unrelated stereotype categories (ICAT moves by ≤ 2 points in the non-gender domains on all four models). This is consistent with the compass's structural locality: a rank-2 plane cannot simultaneously rotate multiple orthogonal bias axes.

5.3 Steering: single-prompt flip behavior

A compass plane is useful only if it can be wielded as a lightweight intervention on real prompts. We run a steering demo on all three models: given 10 contextual prompts (5 per context pole) that naturally continue with the opposing categorical token, we inject

$$v \;=\; \alpha\,\sigma_i\,\cos\theta\,u_i \;+\; \alpha\,\sigma_j\,\sin\theta\,u_j$$

at the final-token residual of the compass layer, with θ set to the compass's fitted opposite-pole angle from Table 1 (so the push adversarially opposes the context bias). The probe on Phi-3 targets the singular/plural compass (*his*/*their*); GPT-2 and Gemma target the gender compass (*he*/*she*). Table 8 reports the two honest metrics — mean logit-diff swing and sign-flip rate — at each model's operating α.

Table 8: Single-head steering results. LD is $\mathrm{logit}(t_+) - \mathrm{logit}(t_-)$; "swing" is mean change from the unsteered baseline toward the target pole on context prompts of the opposite sign. Sign-flip rate counts prompts whose LD crosses zero under the push; top-1 flip is the strict argmax switch. All runs use additive $\alpha\sigma$ injection on 10 prompts.

Model	Layer/Head	α	LD swing $+ \to -$	LD swing $- \to +$	Sign-flip	Top-1 flip
GPT-2 Small	L9H7	20	-3.04	$+2.54$	$4/10$	$1/10$
Phi-3 Mini	L24H10	10	-7.09	$+4.86$	$4/10$	$0/10$
Gemma-2-2B	L21H4	30	-3.68	$+3.15$	$1/10$	$1/10$

All three models steer from a single rank-2 injection. Every model shows a coherent LD swing *against* the context bias and produces at least one genuine top-1 flip from the rank-2 plane alone. The operating α varies because compass amplitude is model-specific (Table 1): Phi-3's compass is strongest and flips LD cleanly at $\alpha = 10$; GPT-2 and Gemma need $\alpha = 20$–30 to overcome their stronger context-driven baselines, beyond which further pushes saturate or degrade fluency without improving the flip count. Top-1 flips are rarer than sign-flips on GPT-2 and Phi-3: the compass overrides the *category vote* but not always the full argmax, consistent with the storage-vs-steering picture from §4.6. Gemma, whose compass doubles as its storage channel, converts its one sign-flip directly into a top-1 flip. Taken together with §5.1, a $2\,d_{\text{model}}$-parameter edit (approximately 6K scalars for Phi-3 Mini) applied at one site produces graded, reversible effects on both individual prompts and population-level bias benchmarks.

6 Discussion

What a compass is and is not. A semantic compass is a causal feature localized to an attention head: its plane is a 2D subspace of the head's write-out space, and injections along it produce a

continuous family of behavior changes parameterized by an angle. A compass is *not* a claim that the head exclusively computes this axis. Higher singular directions in the same head will carry other features; the compass picks out one coherent 2D slice in which a categorical axis is readable and steerable.

Why OV-SVD works as a discovery device. In a transformer block, W_{OV} determines the set of directions the head can add to the residual stream. Its top SVD directions are therefore the candidate "output languages" of the head. When a pair of such directions corresponds to a categorical axis (e.g., masculine vs. feminine), it must correspond to *something the head actually writes*, not just something we can correlate post-hoc. This is the structural reason OV-SVD is a stronger discovery method than probing the residual stream directly: the feature is tied to a specific weight matrix from the start.

Why polar shape differs without implying different mechanisms. Section **??** summarized four axes on which the polar plots can differ (amplitude, singular-value scale, phase, neutral offset). None of them change the underlying fact that the plane is causal and sinusoidally-responsive; they only rescale, rotate, and translate the picture. Future work should standardize polar plots by dividing radii by $A(\alpha)$ so that the three models overlay.

Four-way dials: resolved in §4.7. The two-harmonic fit of a four-token probe distinguishes three distinct plane topologies that are indistinguishable from decode-table inspection alone. Phi-3 L24H17 is a genuine four-way dial ($A_2/A_1 = 0.32$), while L24H28 is two overlapping antipodal axes ($A_2/A_1 \leq 0.06$). A natural direction for future work is to sweep all (u_i, u_j) pairs in a model and classify each plane's dial topology automatically.

Limitations.

1. **Prompt dependence.** Baseline logit separations depend on our choice of three neutral / plus / minus prompts. We chose short, ambiguous prompts to minimize baseline bias; robustness to prompt paraphrase remains to be quantified.
2. **Storage vs. steering locality.** Our head-ablation study (§4.6) shows that in Gemma-2-2B the compass plane is both storage-local (14% of signal, $13.5\times$ over any other head) *and* steering-local, whereas in GPT-2 and Phi-3 the compass is steering-local but not storage-local: the signal is distributed redundantly across heads. This is a substantive scope clarification: compass identifies where a rank-2 causal push lands, not necessarily where the model keeps the category when no push is applied.
3. **Tokenizer constraints.** Phi-3 tokenizes leading-space forms as multi-token; we therefore used bare tokens (`month`, `hour`) which may subtly shift the logit-difference target.
4. **Limited axis breadth.** We demonstrate gender in three models and temporal/entity in one. Sentiment, tense, formality, and animacy remain untested.

Future work. A natural next step is automated compass discovery: scan all (ℓ, h, i, j) tuples and rank by the joint score *amplitude-linearity in* $\alpha \times$ *phase-stability* $\times$ *vocabulary-decoding coherence*. Such a search could surface compasses we currently find by reading decode tables manually. A second direction is using compasses for lightweight steering: a rank-2 angular rotation in one head is a $2d_{\text{model}}$-parameter edit, far sparser than full-vector patching.

Conclusion. The SVD of an attention head's OV matrix is not just a diagnostic object: its top singular planes routinely hold causal, low-dimensional semantic features that can be read off directly from the weights. The resulting *semantic compass* is a simple, architecture-agnostic primitive for mechanistic interpretability that complements both circuit-level and feature-level methods.

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A PROMPT LISTS

All causal sweeps average over three prompts per condition. Prompts are chosen to be short and semantically ambiguous so baseline logit separations reflect category priming rather than syntactic constraint.

Gender (GPT-2, Phi-3, Gemma). Neutral: *"The person said that"*, *"Then they said that"*, *"Afterwards, the speaker said that"*. Masculine-context: *"The man laced up his boots because"*, etc. Feminine-context: *"The woman laced up her boots because"*, etc. Full prompt lists in `investigate_helix_usage_validated.py`.

Temporal (Phi-3 L24H17, tokens *month/hour*). Neutral (3): *"The event took about one"*, *"They waited nearly one"*, *"The project finished within one"*. Plus (month-biasing): *"The rent is due every"*, *"Pregnancy lasts about nine"*, *"The magazine arrives once a"*. Minus (hour-biasing): *"The meeting lasts exactly one"*, *"The drive takes roughly one"*, *"She slept for a single"*.

Entity (Phi-3 L24H28, tokens *item/operation*). Neutral (3): *"The list contained a single"*, *"He removed exactly one"*, *"The inventory showed one"*. Plus (item-biasing): *"On the shelf sat a small"*, *"The box held only one"*, *"She picked up a single"*. Minus (operation-biasing): *"The surgeon completed a complex"*, *"The factory ran one more"*, *"The team executed a risky"*.

B FULL LOGIT-DIFFERENCE CURVES

Figures 5–9 show the raw $\mathrm{LD}(\theta)$ curves at $\alpha \in \{1, 3, 10\}$ for each (model, axis) pair.

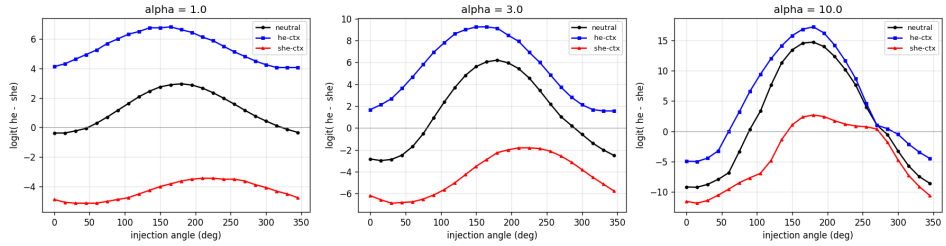


Figure 5: GPT-2 Small, L9H7 gender compass, $\mathrm{LD}(\theta)$.

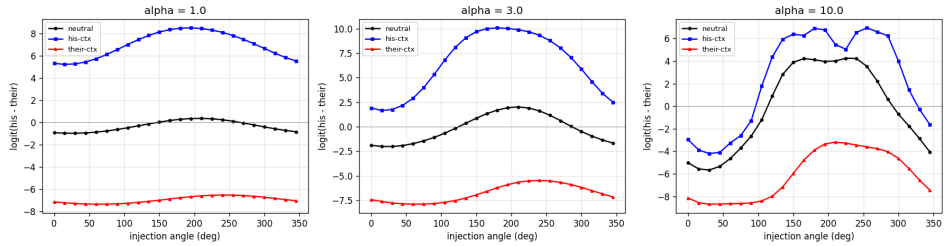


Figure 6: Phi-3 Mini, L24H10 gender compass, $\mathrm{LD}(\theta)$.

C REPRODUCIBILITY

All experiments use the following scripts. Seeds are fixed in the pipeline; the α-sweep is deterministic for fixed prompts.

- `experiments/compass_causal_sweep.py` — causal α-sweep and polar / curves / linearity figures.

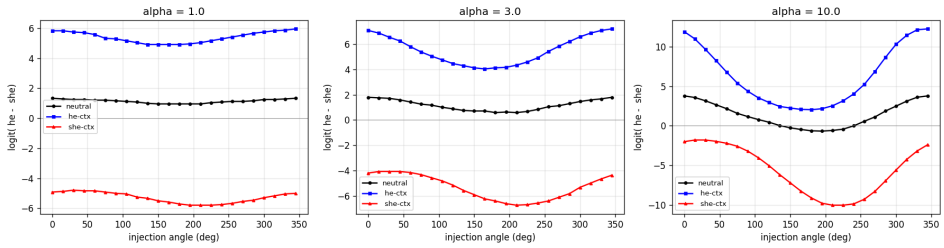


Figure 7: Gemma-2-2B, L21H4 gender compass, LD(θ).

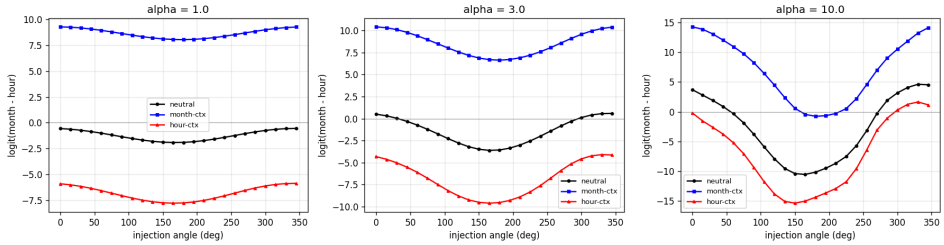


Figure 8: Phi-3 Mini, L24H17 temporal compass, LD(θ).

- `investigate_helix_usage_validated.py` — nine-test battery (`--test decode-heads` through `--self-test`, with `--layer` / `--head` / `--dims` overrides).
- `src/models/online_svd_scanner.py` — head survey and plane enumeration (`analyze_helix`).
- `experiments/baseline_comparison.py` — five-method causal-sweep table (compass, linear probe, ActAdd, contrastive gradient, Fisher top-2) on a shared unit-normalized harness.
- `experiments/head_ablation_comparison.py` — head and plane ablation sweep for the gender axis across all heads in the target layer and matching heads in nearby layers.
- `experiments/fourway_probe.py` — four-token probe with two-harmonic fit for distinguishing two-way vs. four-way compass topology.
- `experiments/steering_demo.py` — rank-2 additive injection on contextual prompts; reports logit-diff swing and sign-flip rate.

All raw outputs are stored in `helix_usage_validated/`; the script `experiments/compass_causal_sweep.py` regenerates the compass figures in `paper_compass/figures/`.

Downstream-bias scripts.

- `experiments/winogender_eval.py` / `winogender_sweep.py` — WinoGender sweep across models and $\alpha \in \{0.25, 0.5, 1.0, 1.5, 2.0\}$.
- `experiments/stereoset_eval.py` — StereoSet LMS/SS/ICAT across domains.
- `experiments/phi3_l28h1_tests.py` — principal-angles and decode tests for the late-layer Phi-3 head.
- `experiments/scan_ci_summary.py` — Agresti-Coull CI aggregation for scan pass rates (Table 5).
- `experiments/run_llama_suite.sh` — end-to-end reproduction of Llama-3.2-3B blind scan, compass, and WinoGender/StereoSet runs.

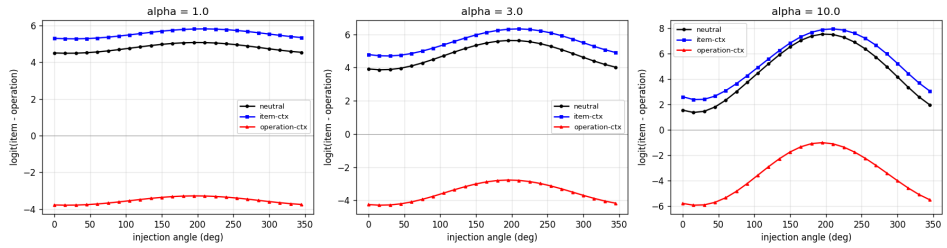


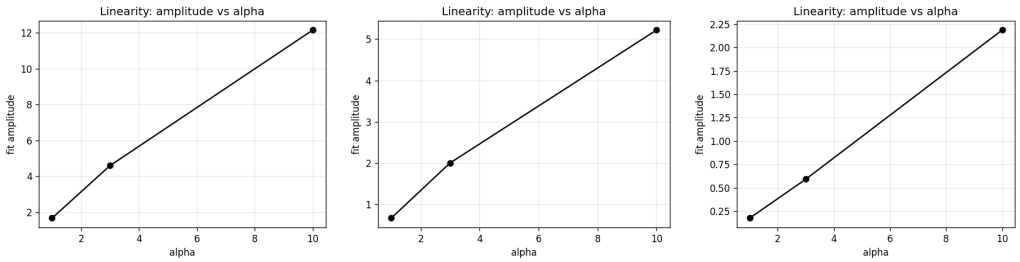
Figure 9: Phi-3 Mini, L24H28 entity compass, LD(θ).

Table 9: Baseline comparison on the gender axis, under equal-push unit-normalized injection. $A(\alpha)$: sinusoidal fit amplitude; slope: $A(10)/10$; $\Delta\phi$: phase drift across $\alpha \in \{1, 3, 10\}$. *Bold* entries are phase-stable ($\Delta\phi < 10°$). Gemma Compass row uses unit-normalized injection; native $\alpha\sigma$ injection (Table 1) recovers $\Delta\phi = 1.3°$ on the same plane.

Model	Method	$A(1)$	$A(3)$	$A(10)$	$\Delta\phi$
GPT-2 Small	Compass	**0.08**	**0.25**	**0.75**	**3.2°**
	Linear probe	**0.14**	**0.41**	**1.29**	**3.4°**
	ActAdd	**0.15**	**0.43**	**1.36**	**3.3°**
	Contrastive grad	**0.23**	**0.69**	**2.23**	**0.4°**
	Fisher top-2	**0.10**	**0.30**	**0.86**	**5.2°**
Phi-3 Mini	Compass	**0.10**	**0.30**	**0.98**	**1.7°**
	Linear probe	0.01	0.02	0.09	23.8°
	ActAdd	0.02	0.06	0.20	22.6°
	Contrastive grad	**0.22**	**0.64**	**2.14**	**0.2°**
	Fisher top-2	0.02	0.07	0.23	342.2°
Gemma-2-2B	Compass	0.07	0.20	0.76	18.1°
	Linear probe	**0.05**	**0.14**	**0.51**	**2.8°**
	ActAdd	**0.06**	**0.16**	**0.61**	**4.2°**
	Contrastive grad	**0.08**	**0.27**	**0.99**	**1.8°**
	Fisher top-2	0.03	0.04	0.20	329.2°

D FULL BASELINE COMPARISON TABLE

E LINEARITY OF AMPLITUDE VS. α



(a) GPT-2 Small, L9H7. (b) Phi-3 Mini, L24H10. (c) Gemma-2-2B, L21H4.

Figure 10: Amplitude $A(\alpha)$ of the fitted sinusoid as a function of injection scale. Linear through the origin $\Rightarrow$ causal plane. L10H9 slope $= 1.42$; L9H7 $= 1.25$; L24H10 $= 0.54$; L21H4 $= 0.22$; L26H14 $= 0.30$; L28H1 $= 1.28$ (values from Table 1). Slope tracks head-specific write strength rather than layer-to-readout depth across models.

Table 10: Two-harmonic fit of four-token probes in Phi-3 at $\alpha \in \{1, 3, 10\}$. $g = \ell_{0^\circ} - \ell_{180^\circ}$ and $h = \ell_{90^\circ} - \ell_{270^\circ}$. Temporal L24H17 h-axis A_2/A_1 grows $0.09 \to 0.20 \to 0.32$ with α, confirming a genuine four-way dial; Entity L24H28 remains ≤ 0.06 throughout.

Compass	α	g A_1	g A_2/A_1	h A_1	h A_2/A_1	Verdict
L24H17 (temporal)	1	0.63	0.03	0.36	0.09	approaching four-way
	3	1.96	0.07	1.02	0.20	four-way
	10	8.06	0.08	3.36	**0.32**	four-way
L24H28 (entity)	1	0.18	0.03	0.17	0.04	two 2-way
	3	0.56	0.01	0.54	0.05	two 2-way
	10	1.90	0.04	1.64	0.06	two 2-way

F TWO-HARMONIC FITS

G FULL SCAN RESULTS

Table 11 lists the top-ranked passing planes from the blind causal scan on Llama-3.2-3B (full table in `helix_usage_validated/llama32_3b_scan_gender_scan.txt`). L26H14 dominates with six of its six top-4 SV-pairs passing; L20H13 adds three near-orthogonal pairs consistent with a second scan site.

Table 11: Top passing planes from the Llama-3.2-3B blind scan. A_{lo}/A_{hi}: amplitude at $\alpha = 3/10$; *slope*: $A_{hi}/10$; $\Delta\phi$: phase drift across α.

Rank	L	H	SVs	A_{lo}	A_{hi}	slope	$\Delta\phi$
1	26	14	(2,3)	3.01	9.36	0.94	2.4°
2	26	14	(1,2)	2.86	9.25	0.93	0.1°
3	26	14	(0,2)	2.85	9.11	0.91	6.4°
4	20	13	(0,1)	2.61	5.14	0.51	0.0°
5	20	13	(0,3)	2.36	4.48	0.45	2.7°
6	20	13	(1,3)	1.40	3.35	0.33	6.7°
7	26	14	(1,3)	0.80	2.98	0.30	0.2°
8	26	14	(0,3)	0.79	2.79	0.28	5.4°
9	16	20	(0,2)	1.00	2.09	0.21	0.6°